5	**(a)**	travel through a vacuum / free space		B1	[1]
	(b)	**(i)** B : name: **microwaves**	wavelength: 10^{-4} to 10^{-1} m	B1	
		C : name: **ultra-violet / UV**	wavelength: 10^{-7} to 10^{-9} m	B1	
		F : name: **X –rays**	wavelength: 10^{-9} to 10^{-12} m	B1	[3]
		(ii) $f = \dfrac{3 \times 10^{8}}{500 \times 10^{-9}}$		C1	
		$f = 6(.0) \times 10^{14}$ Hz		A1	[2]

	(c)	vibrations are in one direction perpendicular to direction of propagation / energy transfer or good sketch showing this	M1 A1	[2]
6	**(a) (i)**	electron	B1	[1]